GIRIDHARAN R

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Bangalore, Karnataka, India

Profile Summary

Information Science Engineering graduate with hands-on experience in **Python, Django, React.js, SQL, Agentic AI and Generative AI**. Experienced in developing full-stack web applications, backend APIs, database-driven systems, and AI-powered applications using **RAG, FAISS, embeddings, and LLM APIs**. Built and deployed projects involving real-time data processing, PDF question answering, fault detection, and personalized learning platforms. Strong foundation in **DSA, OOPs, DBMS, and software development**, with experience using Git, GitHub, Docker, and PostgreSQL.

Technical Skills

Programming Languages: Python, SQL, JavaScript (ES6+)
Backend Development: FastAPI, Django, RESTful APIs, API Integration, Authentication & Authorization
Generative AI & AI Engineering: Generative AI, LLM Integration, Retrieval-Augmented Generation (RAG), Prompt Engineering, Embeddings, Vector Search, Gemini API, LangChain, FAISS
Frontend Development: React.js, HTML5, CSS3, Responsive Web Development
Databases: PostgreSQL, MySQL, SQLite, Database Design, SQL Queries
Tools & Technologies: Git, GitHub, Docker, Vercel, Postman
CS Fundamentals: Data Structures & Algorithms (DSA), Object-Oriented Programming (OOP), DBMS, REST API Design

Projects

AURA X FITNESS Commercial Web Platform & Membership Engine Live Demo
React 19, Vite, JavaScript (ES6+), Vanilla CSS, Schema.org JSON-LD, Google Apps Script, Vercel, DNS

- Architected and deployed a responsive production web platform on a custom domain (`auraxfitness.in`) for a 2,500+ sq.ft. athletic facility, featuring interactive floor previews and a custom glassmorphism design system.
- Developed a tiered membership portal with a Student Verification engine, integrating Google Apps Script webhooks to process base64 ID card uploads and dynamically unlock discounted rates.
- Engineered a One Day Pass booking sub-app supporting live UPI QR code payments, 1-tap mobile intents, automated verification token generation, entry check-in QR codes, and instant WhatsApp receipt dispatch.

AI-Powered PDF Question Answering & RAG System Live Demo
FastAPI, React.js, Gemini API, FAISS, LangChain, PyPDF

- Built a Retrieval-Augmented Generation (RAG) application to answer context-aware questions on uploaded PDF documents.
- Implemented PDF extraction, document chunking, Gemini embeddings, FAISS vector similarity search, and context-grounded Gemini LLM responses.
- Developed REST APIs using FastAPI and integrated a responsive React.js web interface for PDF upload and interactive question answering.

V.E.R.A. – Voice-Enabled EV Intelligence & Personal AI Agent
Python, FastAPI, Gemini, MQTT, PostgreSQL, WebRTC

- Developed a voice-first AI agent for EVs combining LLM-based reasoning, real-time vehicle telemetry, contextual memory, and event-driven automation for personalized driver assistance.
- Implemented persistent and contextual memory with people memory, location-triggered reminders, future messages, routine learning, and proactive "Before You Leave" intelligence.
- Built an event-driven architecture using FastAPI, MQTT, GPS-based triggers, and deterministic Python services for vehicle events, charging states, scheduling, geofencing, and automated actions.

Education

AMC Engineering College, Bangalore 2022 – 2026
B.E. – Information Science and Engineering **CGPA: 7.7/10**

St. Joseph's Pre-University College 2022
PUC **83%**

Certifications

- HackerRank: Python (Basic), SQL (Intermediate), Problem Solving (Intermediate)
- Python Full Stack Development Internship – Dhee Coding Lab (2026)